

Homework 10

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MATH 421 — The Theory of Single Variable Calculus
Section 003
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Collaboration Statement

I worked on this homework independently. No outside collaborators or resources were used.

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Problem 1 (*Ross Chapter 3 18.5 (a)*)

Discussion.

Proof.

□

Problem 2 (*Ross Chapter 3 18.10*)

Discussion.

Proof.

□

Problem 3 (*Ross Chapter 3 19.6 (b)*)

Discussion.

Proof.

□

Problem 4 (*Ross Chapter 3 19.7*)

Discussion.

Proof.

□

Problem 5 (*Uniform Continuity of f*)

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$. Prove that if there exists a real number $M \geq 0$ such that $|f(x) - f(y)| \leq M|x - y|$ for all $x, y \in \mathbb{R}$, then f is uniformly continuous on $\mathbb{R}$.

Discussion.

Proof.

□

Problem 6 (*Differentiability of f*)

Suppose that $0 \in (a, b)$ and $f(x) = xg(x)$ for some function $g : (a, b) \rightarrow \mathbb{R}$ which is continuous at $x = 0$. Prove that f is differentiable at $x = 0$, and find $f'(0)$ in terms of g .

Hint: Note that you cannot use the product rule, since $g(x)$ is not guaranteed to be differentiable at $x = 0$

Discussion.

Proof.

□